

Assignment 3: Virtual Memory Management

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Deadline : 8th November 11:59 PM

Problem Statement

- You are required to implement your own dynamic memory allocation functions: `malloc_*` and `my_free` with the same calling conventions and signatures as the standard `malloc` and `free`. The implementation has to be done in C.
- Each call to a `malloc_*` should return a pointer to the **head of the allocated memory region**. This pointer will be passed as input to `my_free`.
- You must implement an internal **Free Space Manager (FSM)** responsible for handling allocation and deallocation requests.
- Implement a data structure for **free list** to manage available memory blocks.
- Implement the following allocation strategies (each strategy is of 2 marks):
 - (a) **First Fit**
 - (b) **Next Fit**
 - (c) **Best Fit**
 - (d) **Worst Fit**
 - (e) **Buddy Allocation**
- Parts (a) and (b) must be implemented so that any new memory allocation should take time in the order of $\log$ in the number of available free segments. Part (c) and (d) can take linear time.
- Parts (a)-(d) will be called any number of times in any order in a single testcase. Part (e) will be tested separately.
- Use system calls such as `mmap`, `brk`, and `sbrk` for memory allocation. Refer to their respective `man` pages for details.
- Complete the functions provided in the template code and ensure that your implementation behaves identically to the standard memory management library when linked to a user program.
- Each strategy should clearly show how memory blocks are allocated and freed, along with corresponding free list updates.

Submission

- Submit a zip file containing a single header file named `EntryNommu.h` (e.g., `2021MT60986mmu.h`).
- Ensure that your code is well-commented, modular, and adheres to best coding practices for clarity and maintainability.
- All submissions will undergo plagiarism checks against past submissions and AI-generated content. Any form of plagiarism will result in a zero score.

Good luck, and happy coding!